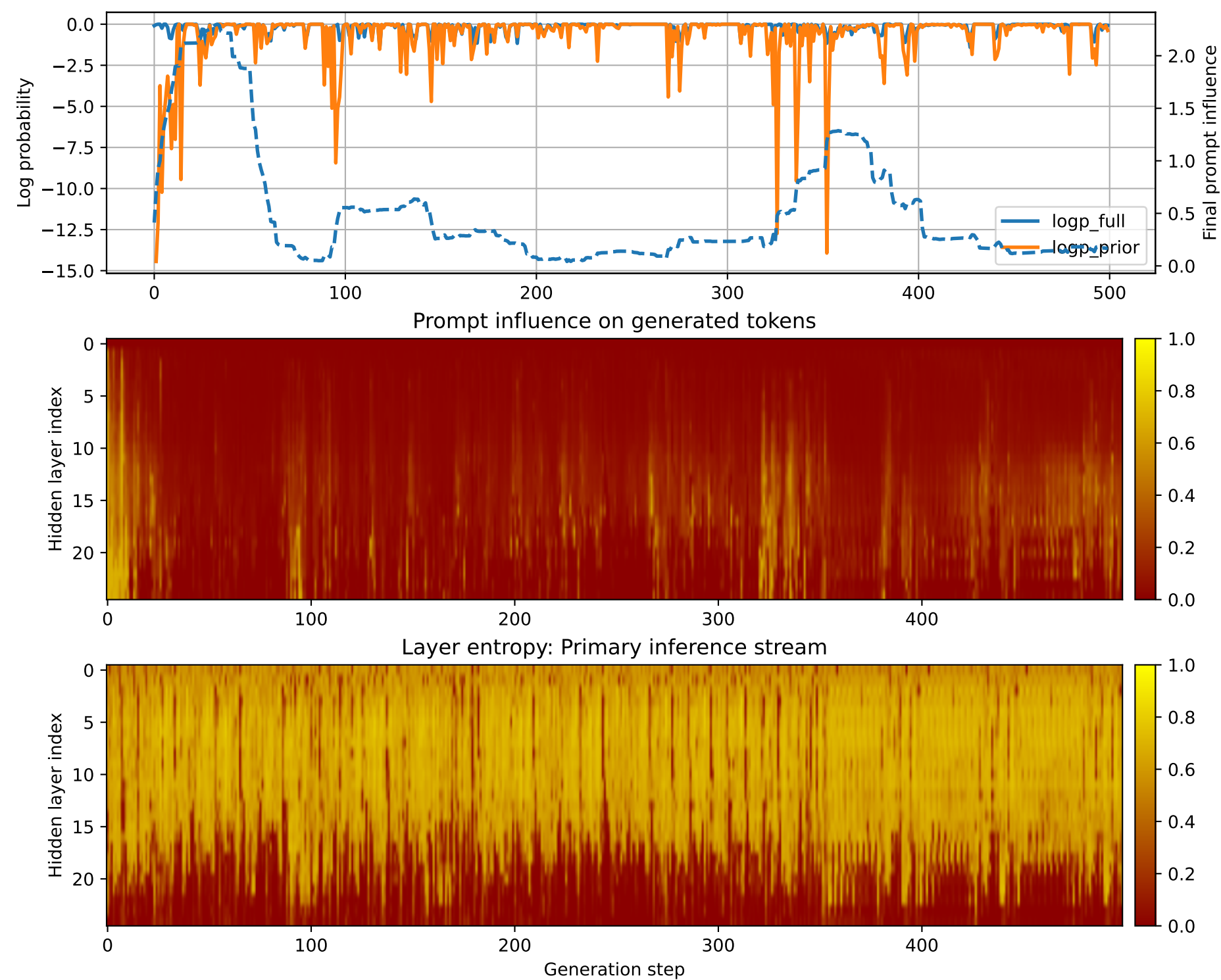


Prompt 1



PROMPT:
Task ID: PYC072 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC072 Your code must contain this required symbol:
def eval_rpn Implement this task: define
function eval_rpn(tokens).. Start your answer with the
check-token line, then provide the Python
implementation.

MODEL RESPONSE:
CHECK_TOKEN: eval_rpn # Define function eval_rpn(tokens) def
eval_rpn(tokens): # Check if the input is
a list of tokens if not isinstance(tokens, list):
raise ValueError("Input must be a list of
tokens.") # Check if the list contains at least one token
if len(tokens) < 1: raise
ValueError("Input must contain at least one token.") #
Define the function to evaluate the RPN (Reverse
Polish Notation) expression def evaluate_rpn(expression):
Split the expression into individual
tokens tokens = expression.split() # Check if
the first token is the operator if
tokens[0] != "#": raise ValueError("First token must
be the operator (e.g., +, -, *, /, %, etc.)")
Split the operator into left and right operands
left_operand = tokens[1] right_operand =
tokens[2:] # Check if the left operand is a number
if not isinstance(left_operand, (int,
float)): raise ValueError("Left operand must be a
number.") # Check if the right operand
is a number if not isinstance(right_operand, (int,
float)): raise ValueError("Right
operand must be a number.") # Evaluate the left operand
using the eval function left_result =
eval(left_operand) # Evaluate the right operand using
the eval function right_result =
eval(right_operand) # Return the result of the
evaluation return left_result + right_result
Evaluate the RPN expression result =
evaluate_rpn(tokens[0]) return result # Test the function
tokens = ["#+", "2", "3", "4", "5", "6", "7", "8", "9", "0"]
print(eval_rpn(tokens)) # Output: 35 tokens = [
"+", "2", "3", "4", "5", "6", "7", "8", "9", "0", "1"]
print(eval_rpn(tokens)) # Output: 15 tokens = [
"+", "2", "3", "4", "5", "6", "7", "8", "9", "0", "1", "2"]
print(eval_rpn(tokens)) # Output: 15 tokens = [
"+",
"2